Travelling fast

10a

Speed up your calculations by using a calculator.

1 Speed is how fast you are going. It is defined as how far you go in a unit of time. **Speed** is calculated by dividing the **distance** travelled by the **time** taken to travel the distance.
Use the bold words above to write a formula for calculating speed.

_____

2 Distance (i.e. how far) is often measured in **metres** or **kilometres**, while time (i.e. how long) is often measured in **seconds** or **hours**.
In the table below write names and abbreviations for the bold units for distance and time.

Distance units		**Time units**	
Name	**Abbreviation**	**Name**	**Abbreviation**

3 Suggest useful units for measuring the speeds involved in the situations listed in the table below.

Situation	**Possible unit**	**Abbreviation**
A car driving on a highway		
A rocket flying through space		
A person running a short race		
A snail crawling along the ground		

4 To compare two speeds, the speeds must have the same unit, such as km/h or m/s.

a A speed of 1 m/s is the same speed as a speed of:

_____ km/h.

b Using the information in the box at right as a model, write an expression for 1 km/h in terms of m/s.

Converting 1 m/s into km/h
Recall: 1 m = 1/1000 km = 0.001 km
1 h = 60 . 60 s = 3600 s; 1 s = 1/3600 h

Substitute 1 m with 1/1000 km and 1 s with 1/3600 h.

1 m/s = 1 . 1/1000 . 3600 m/h = 1 . 3600/1000 km/h
= 3.6 km/h

1 km/h = _____

_____

A speed of 1 km/h equals _____ m/s.

c Express each activity (distance–time relationship) below as a speed, in both m/s and km/h. Show your working out.

Activity	**Calculating speed in m/s**	**Calculating speed in km/h**
Walking 10 km in 2 h		
Jogging 400 m in 60 s		
Running 100 m in 20 s		
Swimming 50 m in 60 s		
Cycling 50 km in 2 h		

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10b

5 Sometimes your speed is known. Knowing speed lets you calculate the time needed to cover a certain distance.

a Write the formula for speed. _____

b Rewrite the formula so that time becomes the subject of the formula.

_____

c Calculate the time needed to cover each distance listed in the table. The speeds to be used for each distance are indicated in the speed column.

Speed	Distance to be covered	Calculation of time needed
School speed limit: 40 km/h	0.2 km past a school (200 m)	
Town speed limit: 60 km/h	2 km along the main road	
Sydney Harbour Bridge speed limit: 70 km/h	500 m across the bridge	
Freeway speed limit: 110 km/h	300 km from Sydney to Canberra	

6 Some careful drivers say a 2 s time gap should be left between moving cars. Obviously this time gap equates to bigger distances as cars travel at higher speeds. When cars are travelling at 30 km/h, what distance is represented by a 2 second gap? To express the 2 second gap in terms of metres, the speed of 30 km/h can be converted to m/s and then used to find the distance the car would travel in 2 s. Recall that 1 km/h = 1/3.6 m/s. Hence 30 km/h = 30 . 1/3.6 m/s = 8.33 m/s. A 2 s time gap corresponds with a distance of 8.33 . 2 m = 16.66 m. This distance divided by the length of a car (i.e. 16.66/4 = 4.17) tells you that you should be more than 4 car lengths behind the car in front of you.

a Write down the formula for speed.

_____

b Rearrange the formula and make distance the subject.

_____

c Now complete the table, showing your calculations as well as the answers. The table shows the time gap measured in seconds, and the distance gap in metres and in car lengths (assuming that a car is 4 m long like a medium-sized car).

Speed (km/h)	Speed (m/s)	Time gap (s)	Distance gap (m)	Distance (4m car lengths)
30	30/3.6 = 8.33	2	8.33 . 2 = 16.66	16.66 / 4 = 4.17
50				
80				
110				

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